

Module-02

Problem Statement:

The following class is a simple class for the operations in a Bank. The objects of Bank class stores id, amount of each client and n keeps track of total clients.

```
class Bank
{
    private:
        int id;
        float amount;
        static int n;
    public:
        //write methods };
```

Now extend the program do the following

- i) Write empty constructor to initialize id and amount to 0
- ii) Write parameterized constructor to initialize id and amount
- iii) Write copy constructor to Initialize id and amount
- iv) Write a method to set id and amount
- v) Write a method to change the amount only
- vi) Write a method to display n only
- vii) Write a method to display id, amount and n where their values can't change.
- viii) Create five clients with id and amount (use all constructors to do it)
- ix) Find total amount of the bank

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define newl cout<<endl
class Bank
{
private:
    int id;
    float amount;
    static int n;
    static int total_balance;
public:
    Bank()
    {
        id=0;
        amount=0;
        n++;
    }
    Bank(int Id, float Amount)
    {
        id=Id;
        amount=Amount;
        total_balance+=Amount;
        n++;
    }
    // copy constructor
```

```

Bank(int a,const Bank&other)
{
    id=a;
    amount=other.amount;
    total_balance+=other.amount;
    n++;
}
void set_info(int i,float at)
{
    id=i;
    amount=at;
    total_balance+=at;
}
void deposit(float Amount)
{
    if(Amount>0)
    {
        amount+=Amount;
        total_balance+=Amount;
        cout<<"You have deposited "<<Amount;
        newl;
    }
    else
    {
        cout<<"Invalid amount";
        newl;
    }
}
void withdraw(float am)
{
    if(amount>=am)
    {
        amount-=am;
        total_balance-=am;
        cout<<"You have withdrawn "<<am;
        newl;
    }
    else
    {
        cout<<"you don't have sufficient balance";
        newl;
    }
}
void Transfer(float am, Bank &other)
{
    if(am<=amount)
    {
        amount-=am;
        other.amount+=am;
        cout<<"Id "<<id<<" has transfered "<<am<<" to Id "<<other.id<<" is successfull\n";
    }
    else
    {
        cout<<"invalid amount\n\n";
    }
}

```

```

}
void get_balance_and_client() const
{
    cout<<"Your id is->";
    cout<<id;
    cout<<" , Your account balance is "<<amount;
    cout<<" , Total Client is->"<<n;
    newl;
}
void Bank_balance()
{
    cout<<"Total balance of bank is ";
    cout<<total_balance;
    newl;
}
};
int Bank::n=0;int Bank::total_balance=0;
int main()
{
    Bank p1;
    Bank p2(2,10000);
    p1.set_info(1,5000);
    Bank p3(3,p1);
    Bank p4(4,12000),p5(5,15000);
    p1.get_balance_and_client();
    p2.get_balance_and_client();
    p3.get_balance_and_client();
    p4.get_balance_and_client();
    p5.get_balance_and_client();
    p1.Bank_balance();
    p5.Transfer(3000,p1);
    p5.get_balance_and_client();
    p1.get_balance_and_client();
    p4.withdraw(1000);
    p4.get_balance_and_client();
    p4.Bank_balance();
    p1.deposit(1500);
    p1.get_balance_and_client();
    p1.Bank_balance();
}

```

Output:

```
Your id is->1, Your account balance is 5000, Total Client is->5
Your id is->2, Your account balance is 10000, Total Client is->5
Your id is->3, Your account balance is 5000, Total Client is->5
Your id is->4, Your account balance is 12000, Total Client is->5
Your id is->5, Your account balance is 15000, Total Client is->5
Total balance of bank is 47000
Id 5 has transfered 3000 to Id 1 is successfull
Your id is->5, Your account balance is 12000, Total Client is->5
Your id is->1, Your account balance is 8000, Total Client is->5
You have withdrawn 1000
Your id is->4, Your account balance is 11000, Total Client is->5
Total balance of bank is 46000
You have deposited 1500
Your id is->1, Your account balance is 9500, Total Client is->5
Total balance of bank is 47500
```

Press any key to continue . . .